

## LYUBAN, Russian Federation

WMO: 260780

Lat: 59.350N

Lon: 31.233E

Elev: 128

StdP: 14.63

Time Zone: 3.00 (E03)

Period: 90-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |             |             |            |             | Coldest Month WS/MCDB |             |             |             | MCWS/PCWD to 99.6% DB |           | WSF        |
|---------------|-------------|-------------|-------------------------------|------------|-------------|-------------|------------|-------------|-----------------------|-------------|-------------|-------------|-----------------------|-----------|------------|
|               |             |             | 99.6%                         |            |             | 99%         |            |             | 0.4%                  |             | 1%          |             |                       |           |            |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB        | DP          | HR         | MCDB        | WS                    | MCDB        | WS          | MCDB        | MCWS                  | PCWD      |            |
| (a)<br>1      | (b)<br>-9.7 | (c)<br>-3.1 | (d)<br>-14.1                  | (e)<br>2.6 | (f)<br>-9.6 | (g)<br>-7.1 | (h)<br>3.8 | (i)<br>-2.2 | (j)<br>15.2           | (k)<br>30.1 | (l)<br>13.6 | (m)<br>28.7 | (n)<br>2.2            | (o)<br>30 | (p)<br>N/A |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 17.2                   | 81.8            | 65.6 | 78.4 | 64.2 | 75.2 | 62.5 | 68.3                | 76.6 | 66.4 | 74.8 | 64.5 | 72.1 | 4.4                  | 180  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 65.6                            | 94.9 | 72.4 | 63.5 | 88.0 | 69.6 | 61.5 | 81.8 | 67.7 | 32.7          | 76.6 | 31.1 | 74.9 | 29.7 | 72.5 | 73.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |
|                   |      |     | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |
| 12.2              | 10.7 | 9.6 | -13.2                      | 84.7 | 8.8                | 3.8 | -19.6                                              | 87.4 | -24.8      | 89.6 | -29.7      | 91.7 | -36.2      | 94.5 |
|                   |      |     | -13.6                      | 70.9 | 8.7                | 1.4 | -19.9                                              | 71.9 | -25.0      | 72.8 | -29.9      | 73.6 | -36.2      | 74.6 |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 41.7   | 20.9  | 24.9  | 29.8  | 40.7  | 52.2  | 58.5  | 62.9  | 60.6  | 51.0  | 39.5  | 30.6  | 27.2  |  |  |
|                                                                        | DBStd       | 16.60  | 11.96 | 10.47 | 8.53  | 7.31  | 7.67  | 6.69  | 5.71  | 5.47  | 6.40  | 7.70  | 9.55  | 9.59  |  |  |
|                                                                        | HDD50       | 4270   | 901   | 703   | 627   | 288   | 64    | 9     | 0     | 1     | 62    | 329   | 581   | 705   |  |  |
|                                                                        | HDD65       | 8602   | 1366  | 1123  | 1092  | 728   | 401   | 213   | 111   | 157   | 421   | 789   | 1031  | 1170  |  |  |
|                                                                        | CDD50       | 1232   | 0     | 0     | 0     | 9     | 133   | 263   | 399   | 330   | 93    | 5     | 0     | 0     |  |  |
|                                                                        | CDD65       | 89     | 0     | 0     | 0     | 0     | 5     | 18    | 44    | 21    | 1     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 931    | 0     | 0     | 0     | 0     | 88    | 213   | 407   | 214   | 9     | 0     | 0     | 0     |  |  |
|                                                                        | CDH80       | 163    | 0     | 0     | 0     | 0     | 20    | 37    | 76    | 28    | 2     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 4.5    | 5.1   | 5.0   | 5.1   | 4.4   | 4.4   | 4.2   | 3.8   | 3.6   | 3.8   | 4.6   | 4.9   | 5.2   |  |  |
|                                                                        | PrecAvg     | 26.8   | 2.1   | 1.5   | 1.4   | 1.7   | 2.3   | 2.8   | 3.5   | 3.0   | 2.3   | 2.2   | 2.4   | 1.9   |  |  |
| (15) Precipitation                                                     | PrecMax     | 37.3   | 3.0   | 2.8   | 2.4   | 3.7   | 5.5   | 4.4   | 9.2   | 5.9   | 4.6   | 3.1   | 3.9   | 3.0   |  |  |
|                                                                        | PrecMin     | 20.5   | 0.8   | 0.1   | 0.4   | 0.6   | 0.8   | 0.9   | 0.9   | 1.0   | 0.7   | 1.0   | 0.1   | 1.0   |  |  |
|                                                                        | PrecStd     | 4.1    | 0.6   | 0.6   | 0.6   | 0.9   | 1.2   | 1.0   | 2.1   | 1.3   | 1.2   | 0.6   | 0.9   | 0.5   |  |  |
|                                                                        | PrecStd     | 4.1    | 0.6   | 0.6   | 0.6   | 0.9   | 1.2   | 1.0   | 2.1   | 1.3   | 1.2   | 0.6   | 0.9   | 0.5   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 40.9  | 43.2  | 53.4  | 69.3  | 82.8  | 84.6  | 85.8  | 83.3  | 73.5  | 59.2  | 47.6  | 46.5  |  |  |
|                                                                        |             | MCWB   | 39.7  | 40.1  | 41.3  | 53.3  | 61.0  | 67.5  | 67.7  | 65.6  | 63.7  | 53.2  | 45.0  | 44.6  |  |  |
|                                                                        | 2%          | DB     | 37.8  | 39.7  | 47.1  | 63.9  | 75.6  | 79.3  | 82.1  | 79.6  | 68.2  | 54.9  | 45.4  | 41.6  |  |  |
|                                                                        |             | MCWB   | 36.1  | 36.9  | 40.0  | 49.8  | 58.8  | 64.3  | 66.4  | 65.4  | 59.8  | 50.6  | 43.6  | 40.0  |  |  |
|                                                                        | 5%          | DB     | 36.0  | 37.4  | 42.9  | 59.3  | 71.9  | 75.8  | 78.8  | 76.1  | 64.5  | 52.4  | 43.2  | 39.3  |  |  |
|                                                                        |             | MCWB   | 34.6  | 35.3  | 37.4  | 47.7  | 57.9  | 62.1  | 64.8  | 64.0  | 57.5  | 49.7  | 41.8  | 38.2  |  |  |
|                                                                        | 10%         | DB     | 34.5  | 35.7  | 40.3  | 54.7  | 67.6  | 72.4  | 75.0  | 72.3  | 61.6  | 50.5  | 41.2  | 37.1  |  |  |
|                                                                        |             | MCWB   | 33.3  | 34.0  | 36.2  | 45.8  | 55.5  | 60.1  | 63.9  | 62.1  | 55.9  | 48.2  | 40.0  | 35.9  |  |  |
|                                                                        | 0.4%        | WB     | 40.0  | 40.8  | 43.7  | 54.1  | 65.6  | 70.9  | 71.3  | 69.9  | 65.1  | 54.0  | 45.7  | 44.1  |  |  |
|                                                                        |             | MCDB   | 41.0  | 43.0  | 48.9  | 65.2  | 75.1  | 81.0  | 79.8  | 79.3  | 72.8  | 57.1  | 47.0  | 46.4  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 36.6  | 37.6  | 41.0  | 51.6  | 61.4  | 66.9  | 68.9  | 66.7  | 61.2  | 52.0  | 43.9  | 40.2  |  |  |
|                                                                        |             | MCDB   | 37.5  | 39.2  | 46.2  | 60.9  | 70.7  | 75.1  | 76.6  | 75.1  | 66.4  | 53.8  | 45.1  | 41.3  |  |  |
|                                                                        | 5%          | WB     | 34.8  | 35.6  | 38.6  | 49.0  | 59.2  | 63.7  | 67.1  | 65.0  | 58.6  | 50.4  | 42.0  | 38.2  |  |  |
|                                                                        |             | MCDB   | 35.7  | 37.0  | 42.0  | 57.5  | 68.7  | 72.2  | 75.3  | 72.7  | 62.8  | 52.2  | 43.0  | 39.2  |  |  |
|                                                                        | 10%         | WB     | 33.5  | 34.2  | 36.7  | 46.3  | 57.0  | 61.5  | 65.3  | 63.3  | 56.5  | 48.2  | 40.1  | 36.0  |  |  |
|                                                                        |             | MCDB   | 34.5  | 35.3  | 39.6  | 53.6  | 66.7  | 70.2  | 73.1  | 70.3  | 60.5  | 50.2  | 41.1  | 36.9  |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.5   | 9.7   | 13.3  | 17.7  | 20.2  | 18.5  | 17.2  | 17.1  | 14.2  | 10.2  | 6.9   | 7.0   |  |  |
|                                                                        |             | MCDBR  | 6.9   | 8.4   | 16.1  | 27.5  | 27.6  | 25.0  | 24.3  | 24.2  | 20.6  | 13.6  | 7.7   | 6.9   |  |  |
|                                                                        |             | MCWBR  | 6.6   | 6.9   | 10.4  | 15.6  | 13.1  | 11.4  | 10.8  | 11.8  | 12.6  | 10.0  | 7.0   | 6.4   |  |  |
|                                                                        |             | MCDBR  | 6.6   | 7.3   | 13.0  | 24.1  | 22.3  | 21.1  | 20.2  | 20.0  | 17.1  | 11.9  | 7.3   | 6.9   |  |  |
|                                                                        | 5% WB       | MCWBR  | 6.5   | 6.6   | 9.4   | 14.5  | 11.7  | 10.8  | 10.4  | 10.7  | 11.3  | 9.6   | 6.8   | 6.5   |  |  |
|                                                                        |             | MCWBR  | 6.5   | 6.6   | 9.4   | 14.5  | 11.7  | 10.8  | 10.4  | 10.7  | 11.3  | 9.6   | 6.8   | 6.5   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.276 | 0.314 | 0.337 | 0.379 | 0.372 | 0.368 | 0.394 | 0.390 | 0.359 | 0.336 | 0.307 | 0.264 |  |  |
|                                                                        | taud        |        | 2.105 | 2.107 | 2.165 | 2.275 | 2.379 | 2.415 | 2.361 | 2.360 | 2.432 | 2.380 | 2.213 | 2.086 |  |  |
|                                                                        | Ebn at Noon |        | 159   | 213   | 249   | 260   | 272   | 275   | 263   | 254   | 244   | 210   | 148   | 119   |  |  |
|                                                                        | Edh at Noon |        | 15    | 26    | 33    | 35    | 34    | 34    | 35    | 32    | 26    | 20    | 14    | 10    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 93    | 290   | 742   | 1173  | 1609  | 1690  | 1621  | 1280  | 786   | 322   | 94    | 51    |  |  |
|                                                                        | RadStd      |        | 18    | 64    | 97    | 150   | 156   | 135   | 198   | 133   | 76    | 52    | 12    | 8     |  |  |

## Historical Trends

|                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                        |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| Station Only           | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| Regional (6 neighbors) | +0.94 | N/A     | N/A    | N/A     | N/A   | N/A   | -328        | -351  | N/A   | N/A   |

Nomenclature: See separate page